

Set 10: Survey Responses Analysis

DSA – Data Handling and Visualization | Lab Experiment

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Aim

To analyse survey answers using a grouped bar chart, a stacked bar chart and a labelled table, and to combine them in an interactive Tableau dashboard.

Software / Tools Used

R 4.3.3 with ggplot2 (charts), gridExtra (tables) and patchwork (dashboard layout); Tableau Desktop / Tableau Public (interactive dashboard).

Dataset

Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

Program – Dataset Creation (common to all R tasks)

```
library(ggplot2)
library(gridExtra)
survey <- data.frame(
  Survey_ID = c(1, 2, 3),
  Question_1 = c("A", "B", "C"),
  Question_2 = c("B", "A", "A"),
  Question_3 = c("C", "D", "B")
)
write.csv(survey, "Survey_Responses.csv", row.names = FALSE) # for Tableau

# Long format: one row per (survey, question, answer)
survey_long <- reshape(survey, direction = "long",
  varying = c("Question_1", "Question_2", "Question_3"),
  v.names = "Answer", timevar = "Question",
  times = c("Question 1", "Question 2", "Question 3"),
  idvar = "Survey_ID")
survey_long$Answer <- factor(survey_long$Answer, levels = c("A", "B", "C", "D"))
counts <- as.data.frame(table(Question = survey_long$Question,
  Answer = survey_long$Answer))
names(counts)[3] <- "Responses"
pal <- c(A = "#0072B2", B = "#E69F00", C = "#009E73", D = "#CC79A7")
```

Q1. Grouped Bar Chart – Distribution of Answers for Question 1

Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.

R Program:

```
# Q1. Grouped bar chart - distribution of answers (Question 1 highlighted)
cat("Answer distribution for Question 1:\n")
print(table(Answer = factor(survey$Question_1, levels = c("A", "B", "C", "D"))))
cat("\nAnswer counts for all questions:\n")
print(xtabs(Responses ~ Question + Answer, data = counts))

counts$Highlight <- ifelse(counts$Question == "Question 1", "Question 1", "Other questions")
p_grp <- ggplot(counts, aes(x = Answer, y = Responses, fill = Question, group = Question)) +
  geom_col(position = position_dodge(width = 0.8), width = 0.75,
    aes(alpha = Highlight), colour = "grey30") +
  geom_text(aes(label = Responses,
    position = position_dodge(width = 0.8), vjust = -0.4, size = 3.8) +
```

```

scale_fill_manual(values = c("Question 1" = "#0072B2", "Question 2" = "#E69F00",
                             "Question 3" = "#009E73")) +
scale_alpha_manual(values = c("Question 1" = 1, "Other questions" = 0.45), guide = "none") +
scale_y_continuous(limits = c(0, 2.5), breaks = 0:2) +
labs(title = "Distribution of Answers (Question 1 highlighted)",
     x = "Answer Option", y = "Number of Responses", fill = "Question") +
theme_minimal(base_size = 12) + theme(plot.title = element_text(face = "bold"))
print(p_grp)
ggsave("s10_q1_grouped.png", p_grp, width = 7.5, height = 4.5, dpi = 150, bg = "white")

```

Console Output:

```

Answer distribution for Question 1:
Answer
A B C D
1 1 1 0

Answer counts for all questions:
      Answer
Question  A B C D
Question 1 1 1 1 0
Question 2 2 1 0 0
Question 3 0 1 1 1

```

Output:

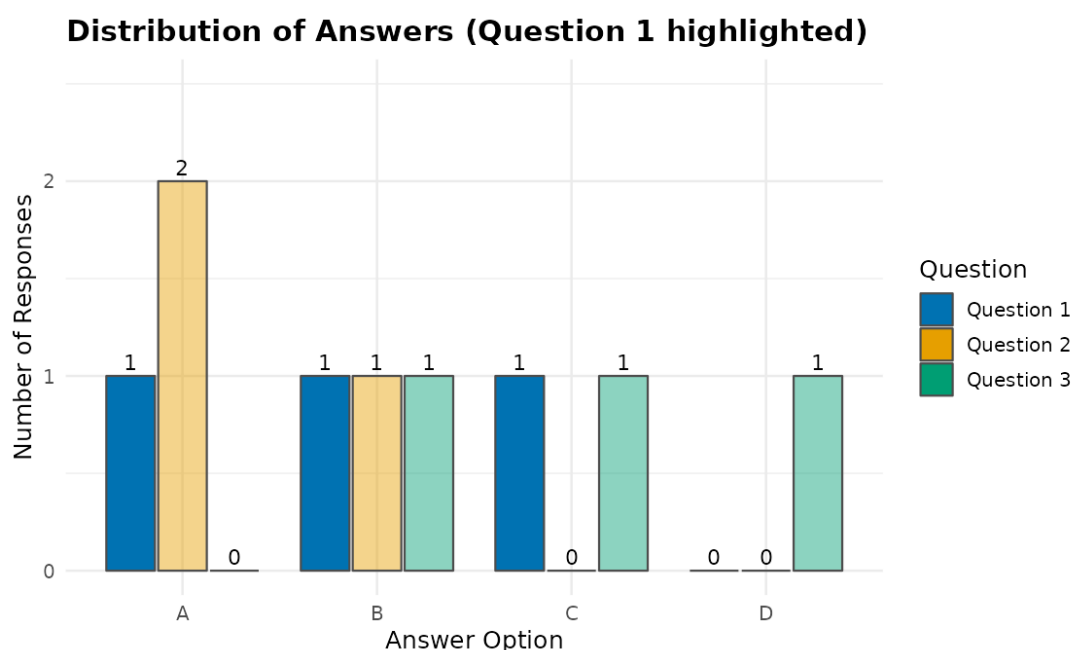


Figure 1: Grouped bar chart of answer options; Question 1 shown in solid blue

Inference:

For **Question 1**, answers A, B and C were each chosen once and D was never chosen, so responses are evenly split with no consensus. Grouping it beside Questions 2 and 3 adds contrast: Question 2 has a clear majority for A (2 of 3).

Q2. Stacked Bar Chart – Overall Distribution for All Questions

Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

R Program:

```

# Q2. Stacked bar chart - overall distribution of responses for all three questions
cat("Overall answer totals across all questions:\n")
print(table(survey_long$Answer))

p_stk <- ggplot(counts[counts$Responses > 0, ],
               aes(x = Question, y = Responses, fill = Answer)) +
  geom_col(width = 0.55, colour = "white") +
  geom_text(aes(label = Answer, position = position_stack(vjust = 0.5),
               colour = "white", fontface = "bold", size = 4.5) +
  scale_fill_manual(values = pal, drop = FALSE) +
  labs(title = "Overall Distribution of Responses by Question",
       x = "Question", y = "Number of Responses", fill = "Answer") +

```

```
theme_minimal(base_size = 12) + theme(plot.title = element_text(face = "bold"))
print(p_stk)
ggsave("s10_q2_stacked.png", p_stk, width = 7, height = 4.5, dpi = 150, bg = "white")
```

Console Output:

Overall answer totals across all questions:

```
A B C D
3 3 2 1
```

Output:

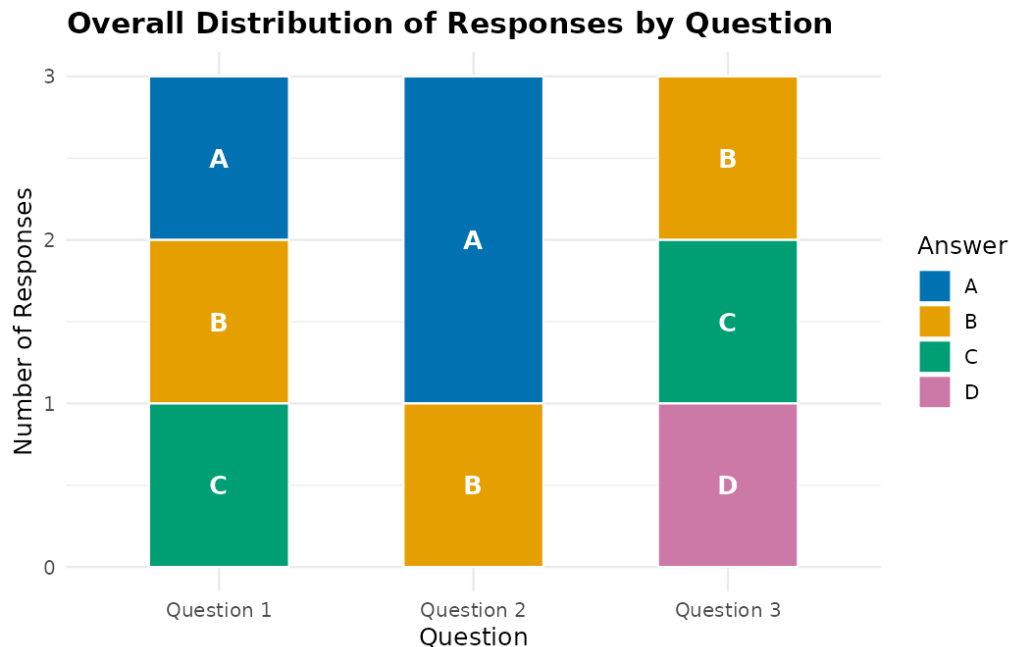


Figure 2: Stacked bar chart of responses for each question

Inference:

Overall, A and B are the most common answers (3 each), C is next (2) and D is least common (1, only in Question 3). Question 2 is the most one-sided (A = 2), while Questions 1 and 3 have their answers spread across three options.

Q3. Table – Survey Response Data

Build a table to show the survey response data for each survey. Label the table elements.

R Program:

```
# Q3. Labelled table of survey responses
tbl <- survey
names(tbl) <- c("Survey ID", "Question 1", "Question 2", "Question 3")
print(tbl, row.names = FALSE)
tg <- tableGrob(tbl, rows = NULL, theme = ttheme_default(
  core = list(fg_params = list(fontsize = 11)),
  colhead = list(fg_params = list(fontsize = 12, fontface = "bold", col = "white"),
    bg_params = list(fill = "#1F3864"))))
title <- grid::textGrob("Survey Response Data",
  gp = grid::gpar(fontsize = 14, fontface = "bold"))
png("s10_q3_table.png", width = 900, height = 330, res = 150)
grid.arrange(title, tg, ncol = 1, heights = c(0.25, 1))
invisible(dev.off())
```

Console Output:

```
Survey ID Question 1 Question 2 Question 3
1         A         B         C
2         B         A         D
3         C         A         B
```

Output:

Survey Response Data

Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

Figure 3: Labelled survey response table

Inference:

The table shows each respondent's complete answer set. No two surveys gave the same answer pattern.

Q4. Tableau Dashboard – Interactive Exploration

Develop a Tableau dashboard combining the grouped bar chart, stacked bar chart, and the table for interactive exploration of survey responses.

R Program:

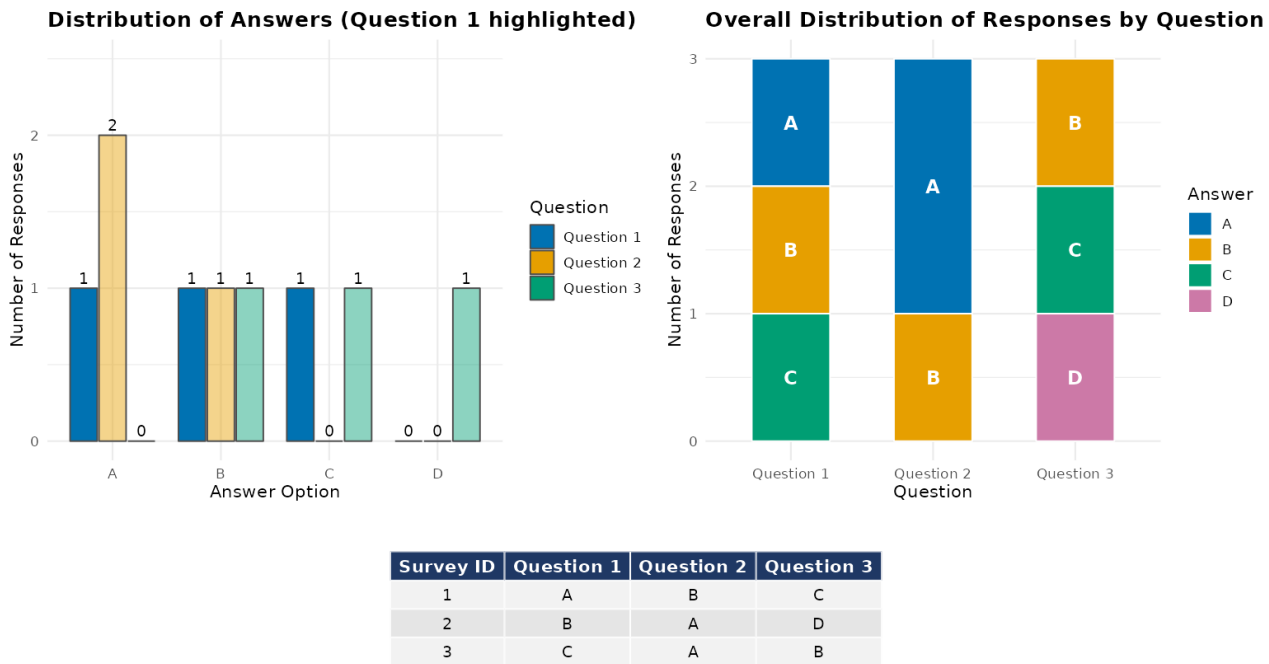
```
# Q4. Dashboard layout preview (Tableau dashboard built from Survey_Responses.csv)
library(patchwork)
dash <- (p_grp | p_stk) / wrap_elements(full = tg) +
  plot_layout(heights = c(1.4, 0.6)) +
  plot_annotation(title = "Survey Responses Dashboard",
                  theme = theme(plot.title = element_text(face = "bold", size = 16)))
ggsave("s10_dashboard.png", dash, width = 12, height = 7, dpi = 130, bg = "white")
```

Procedure in Tableau:

- 1 Open Tableau → **Connect** → **Text file** → **Survey_Responses.csv**. In the Data Source tab, select the Question_1, Question_2 and Question_3 columns → right-click → **Pivot**. Rename *Pivot Field Names* to **Question** and *Pivot Field Values* to **Answer**.
- 2 **Sheet 1 – "Grouped Bar"**: drag Answer to Columns, Question to Columns (to the right of Answer) and **CNT(Survey ID)** to Rows. Drag Question to Color and show labels. Add a Question filter defaulted to **Question 1** (or highlight it) to focus on Question 1.
- 3 **Sheet 2 – "Stacked Bar"**: drag Question to Columns and CNT(Survey ID) to Rows, then drag Answer to **Color**. Bars stack automatically; add Answer to Label.
- 4 **Sheet 3 – "Response Table"**: drag Survey ID and Question to Rows and Answer to Text (or use the original un-pivoted columns as a crosstab). Rename the headers.
- 5 From the menu choose **Dashboard** → **New Dashboard**. Set **Size** → **Automatic** (or Fixed 1200 × 800).
- 6 Drag the sheets onto the canvas: Grouped Bar (top-left), Stacked Bar (top-right), Response Table (bottom). Add a **Text** object at the top for the dashboard title.
- 7 On each chart, click the sheet's drop-down and select **Use as Filter**. Clicking a bar or slice then filters the other views.
- 8 Go to **Dashboard** → **Actions** → **Add Action** → **Highlight** (Run on: Hover) so related marks highlight across all views.
- 9 Right-click a filter → **Apply to Worksheets** → **All Using This Data Source** (Survey_Responses) so one filter controls every view.
- 10 Save as a packaged workbook (**File** → **Save As** → **.twbx**) or publish to Tableau Public.

Dashboard Layout:

Survey Responses Dashboard



Reference layout of the dashboard (rendered in R). The interactive version is built in Tableau using the procedure above.

Inference:

Clicking an answer segment (for example, A) in the stacked bar highlights the surveys in the table that gave that answer, so each response can be traced back to its respondent.

Result

The survey responses were visualised with grouped and stacked bar charts and a labelled table in R, and combined into an interactive Tableau dashboard (using a pivot to reshape the data). Question 1 has an even split across A, B and C; Question 2 has a majority for A; and A and B are the most frequent answers overall.